Set Up A R.A.I.D.

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RAID (Redundant Array of Inexpensive Disk) is the best form of fault tolerance that enables swift data recovery after disasters such as hard drive crashes. You can also use RAID to boost system performance. There are five RAID architectures-RAID-1 to RAID-5-each offering different features and performance trade-offs. 'Mirroring' and 'striping' are two ways of implementing RAID. Mirroring offers data-redundancy while striping offers performance boost.

WAYS TO SETUP RAID

Implement RAID on your server via hardware or software. Software-based RAID is OS-dependent. A few OSes such as Windows 2000 Server support software RAID. No such dependencies plague hardware RAID and since this setup is isolated, it has its own dedicated controller and memory to work without much interference from the server's CPU.

Hardware RAID Setup

Use a PCI RAID card to connect all the hard drives. Unlike software RAID, here the OS can reside on the RAID hard drives, since the RAID volume boots from the RAID card which occurs prior to the OS boot.

RAID cards are available from vendors such as LSI, Adaptec, etc. Depending on your RAID controller card, the interface for creating RAID volumes differs. However, the general idea is the same: first, boot into the adapter card's BIOS by pressing the appropriate key and then create the required volumes on the drive. Next, save the settings, exit the BIOS, and load the OS on one of the volumes. For this workshop, we use the onboard RAID controller on the MSI 875PNEO motherboard.

Note: Provide the adapter card driver the moment you begin installing the OS on a RAID volume. Press [F6] when Windows 2000 starts up to install them. Insert the driver floppy when prompted to do so and follow the instructions thereof.

STEP 1 Once the system boots, press [Ctrl]+[I]. (the key combination defers from card to card. Refer to the manual for the correct sequence) to boot into the RAID card BIOS.

STEP 2 Select 'Create RAID volume' in the interface that comes up. Enter the name for the volume and press [Enter].

Access the RAID BIOS setup screen as soon as the system boots up

STEP 3 Next, using the arrow keys, select the type of volume i.e. RAID 0 or RAID 1. Here, we select RAID 1 and press [Enter].

STEP 4 Decide upon the amount of space you want to allocate to the volume, let's say 20 GB, press [Enter] and confirm it with a 'Y'.

STEP 5 Create another volume (by repeating steps 2 to 4) if there is space left to do so. Once the volumes are created, exit the BIOS and restart the system.

STEP 6 Now install the OS and the required drivers. After installation is complete, you will need to install the RAID management software that comes bundled with most cards.

Software RAID Setup

Use Windows 2000 Server editions to implement software RAID on you server. Linux distributions, such as SuSe, etc, also offer software RAID. Here, though we stick to the Windows platform.

STEP 1 Once the hard drives are installed, initialize the new drives to dynamic drives. To convert your disk to a dynamic drive, right-click 'My Computer' and select 'Manage'.

Note: System drives cannot be converted to dynamic disk.

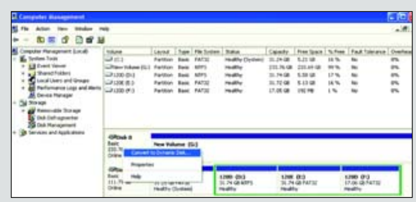
STEP 2 In the 'Computer Management' console that opens, click on 'Disk management'. The installed drives appear in the window pane to the right. Right-click on the new hard drives (not on the volumes) and click the option 'Convert to dynamic disk'.

Glossary

Mirroring (RAID-1) In mirroring, the same data is written simultaneously to both the hard drives. During a hard drive crash, a copy of your data is always present on either of the drives, which can be later retrieved. Hence, mirroring provides data redundancy.

Striping (RAID-0) In striping, alternate chunks of data are written to the two hard drives, i.e. chunk 1 goes to drive A, chunk 2 goes to drive B and so on. Since each drive has its own controller, the overall writing and reading process is faster than what can be achieved using a single controller. Hence, use striping for higher data throughput, eg, file-servers, etc.

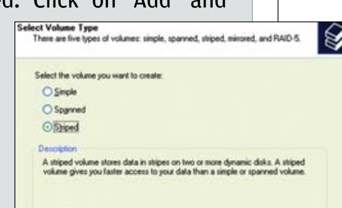
RAID-5 The RAID-5 configuration is an amalgamation of RAID-1 and RAID-0. RAID-5 requires 4 hard drives, one pair for striping and the other to mirror the data. Recent advancements have made it possible to have RAID-5 configuration on two drives.



Select the hard drive and convert them to dynamic disks

STEP 3 Once the disks are converted to dynamic disks, select the empty volumes on the drives and then on 'New volume'. In the New volume wizard, click 'Next'. Depending on your preference, choose 'Striped' or 'Mirrored' volumes and click 'Next'.

STEP 4 In the next window, select the drive volumes to be mirrored or striped. Click on 'Add' and then click 'Next'. Choose the type of file system, check the 'perform quick format' checkbox and click 'Next' to complete the process. Upon completion, the RAID volume shows up with a different colour in the 'Disk management' console.



Select the RAID type in this screen

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